

Statistical & Numerical Methods HW1

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Technion

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1 Poker Game

A pack of cards has 52 cards, 13 of each kind. At the start of the game the pack is shuffled and each player gets 5 cards. We want to know the probability of

1. Have a 2 of clubs in your hand
2. Have a single pair of some number
3. Have a full house
4. Have a flush

I'm assuming there's only one player because otherwise the probabilities depend on the number of players

1.1 2 of clubs

There is only one 2 of clubs and you draw 5 cards so you have 5 opportunities to get the card.

$$P_{tot} = \binom{5}{1} \cdot P_1 = 5 \cdot \frac{1}{52} \approx 9.62\% \quad (1)$$

1.2 Pair

The chance to have a pair of numbers is the number of ways we can draw a number, multiplied by the number of ways we could draw the same number, multiplied by the number of ways the rest of the cards could be drawn to be something else, multiplied by the number of ways we could draw them so it's only one of each, all divided by the total number of ways to draw cards

$$P_{pair} = \frac{\binom{10}{1} \cdot \binom{4}{2} \cdot \binom{12}{3} \cdot \binom{4}{1}^3}{\binom{52}{5}} \approx 32.5\% \quad (2)$$

1.3 Full House

A full house is 3 cards of one rank and 2 of another.

$$P_{fh} = \frac{\binom{13}{1} \binom{4}{3} \binom{12}{1} \binom{4}{2}}{\binom{52}{5}} \approx 0.17\% \quad (3)$$

1.4 Flush

A flush is 5 cards of the same color. I'm assuming royal flushes and straight flushes are also considered a flush.

$$P_f = \frac{\binom{4}{1} \binom{13}{5}}{\binom{52}{5}} \approx 0.19\% \quad (4)$$

2 American Exam

Assume Danny solves an american exam where each question has 5 answers and only one is correct. If Danny knows the correct answer he will choose it, otherwise he guesses at random between the five options. Danny knows the answer for 7 out of 10 questions.

1. For one question in the exam, what is the probability Danny will answer it right?
2. If Danny answered right, what's the probability he guessed it?
3. For an exam with 30 questions, what's the probability Danny will get 100?

2.1 Single Question

$$P(\text{Correct}) = P(\text{Correct}|\text{Knew})P(\text{Knew}) + P(\text{Correct}|\text{Guess})P(\text{Guess}) \quad (5)$$

$$= 1 \cdot \frac{7}{10} + \frac{1}{5} \cdot \frac{3}{10} = \frac{7}{10} + \frac{3}{50} = \frac{19}{25} \quad (6)$$

2.2 Chance of Guess

$$P(\text{guess}|\text{correct}) = \frac{P(\text{correct}|\text{guess})P(\text{guess})}{P(\text{correct})} = \frac{0.2 \cdot 0.3}{\frac{19}{25}} = \frac{3}{38} \quad (7)$$

2.3 Chance of 100

$$P(100) = P(\text{correct})^{30} = \left(\frac{19}{25}\right)^{30} \approx 0.00026 \quad (8)$$

3 Random Coin

$k + 1$ coins are arranged in a row so that the i th coin has a probability of $\frac{i}{k}$ to fall on heads. A coin is selected from the row at random and is flipped n times. If all n flips were heads, what's the chance the $n + 1$ th flip will be heads?

The probability of getting $n + 1$ heads given that we have n heads is given by

$$P(H_{n+1}|H_n) = \frac{P(H_{n+1})}{P(H_n)} \quad (9)$$

since each flip is independent of the others. But

$$P(H_n) = \frac{1}{k+1} \sum_{i=0}^{k+1} \left(\frac{i}{k}\right)^n \quad P(H_{n+1}) = \frac{1}{k+1} \sum_{i=0}^{k+1} \left(\frac{i}{k}\right)^{n+1} \quad (10)$$

so

$$P(H_{n+1}|H_n) = \frac{\sum_{i=0}^{k+1} \left(\frac{i}{k}\right)^{n+1}}{\sum_{i=0}^{k+1} \left(\frac{i}{k}\right)^n} \quad (11)$$

using the hint

$$P(H_{n+1}|H_n) = \frac{\frac{k}{n+2}}{\frac{k}{n+1}} \quad (12)$$

$$= \frac{n+1}{n+2} \quad (13)$$

4 Random Coin

1. Assume two independent random variables x_1, x_2 with variances σ_1^2, σ_2^2 . Calculate

$$\text{cov}(x_1 + x_2, x_1 - x_2) \quad (14)$$

2. Assume a poisson random variable with $p < \lambda < 1$. Calculate $E[x!]$.
3. Assume x, y are independent random variables with the same probability of receiving the values $1, 2, \dots, m$ with the same probability. Show that

$$E[|x - y|] = \frac{(m-1) \cdot (m+1)}{3m} \quad (15)$$

4.1 Covariance

$$\text{cov}(x_1 + x_2, x_1 - x_2) = \text{cov}(x_1, x_1) - \text{cov}(x_1, x_2) + \text{cov}(x_2, x_1) - \text{cov}(x_2, x_2) \quad (16)$$

$$= \sigma_1^2 - \sigma_2^2 - 0 + 0 \quad (17)$$

$$= \sigma_1^2 - \sigma_2^2 \quad (18)$$

4.2 Expectation of Factorial

The expected value is defined by

$$E[x!] = \sum_{k=0}^{\infty} k! P(x = k) \quad (19)$$

and since x is poisson

$$P(x = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad (20)$$

then

$$E[x!] = \sum_{k=0}^{\infty} k! \frac{\lambda^k e^{-\lambda}}{k!} = \sum_{k=0}^{\infty} \lambda^k e^{-\lambda} = \frac{e^{-\lambda}}{1 - \lambda} \quad (21)$$

4.3 Expected Value of Difference

By definition

$$E[|x - y|] = \frac{1}{m^2} \sum_{x=1}^m \sum_{y=1}^m |x - y| \quad (22)$$

to advance we need to look at this from the perspective of summing over the different values $|x - y|$ can give which we'll call d , so its one sum. We notice that $d = 0$ gives no contribution and the maximum difference is $m - 1$. We also notice that for each value of d , there are $2(m - d)$ pairs which give that value of d (m is intuitive since the higher the number the more pairs there are, $-d$ is because as the gap between the

numbers grows, it "takes up space", and the 2 is because the pairs are ordered). So,

$$E[|x - y|] = \frac{1}{m^2} \sum_{d=1}^{m-1} 2d(m-d) \quad (23)$$

$$= \frac{2}{m^2} \sum_{d=1}^{m-1} (md - d^2) \quad (24)$$

$$= \frac{2}{m^2} \left(m \sum_{d=1}^{m-1} d - \sum_{d=1}^{m-1} d^2 \right) \quad (25)$$

using the hint

$$E[|x - y|] = \frac{2}{m^2} \left(\frac{m^2(m+1)}{2} - \frac{m(m+1)(2m+1)}{6} \right) \quad (26)$$

$$= m + 1 - \frac{(m+1)(2m+1)}{3m} \quad (27)$$

$$= m + 1 - \frac{2m^2 + 3m + 1}{3m} \quad (28)$$

$$= m - \frac{2m^2 + 1}{3m} \quad (29)$$

$$= \frac{3m^2 - 2m^2 - 1}{3m} \quad (30)$$

$$= \frac{m^2 - 1}{3m} \quad (31)$$

$$= \frac{(m-1)(m+1)}{3m} \quad (32)$$